

Position Auctions in AI-Generated Content

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Abstract

We consider an extension to classic position auctions in which sponsored creatives are embedded within AI-generated content rather than shown in predefined slots. Leveraging advanced LLM technologies, it becomes viable to seamlessly integrate sponsored creatives with AI content and accurately estimate the context-aware benefits of differing insertion positions. However, this approach introduces novel challenges; substitution effects require rigorous treatment compared to standard position auction settings, where slots are independent of each other.

In this work, we formalize a mathematical model of the extended position auction problem and study the welfare- and revenue-maximization mechanism design problems. We assume a specific click-through rate estimation for each position-creative pair and consider two user behavior models: a multinomial logit (MNL) and a cascade model. For the MNL model, which is order-insensitive, we efficiently implement optimal mechanisms. For the cascade model, which is order-sensitive, we provide approximately optimal mechanisms.

CCS Concepts

• **Theory of computation** → **Algorithmic mechanism design**; **Computational advertising theory**; • **Information systems** → **Language models**.

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1 Introduction

Position auctions [40], the foundational mechanism for sponsored search, have facilitated trillions of dollars in business over the past decades. These auctions sell positions that are identical except for their rank. Under the commonly used separability assumption,¹ the competition for each position becomes effectively identical, leading to a uniform preference order among buyers and a significantly simplified optimization problem.

Inspired by advances in generative AI, we consider an extended position auction setting in which buyers compete for the placement of sponsored creatives within AI-generated commercial content. For example, Microsoft is testing embedding ads in the new Bing generative AI search engine [35] and Perplexity started to show ads inside their generative AI chatbot [36]. See Figure 1 for an example illustration.

This novel application introduces context as a significant factor influencing click-through rates. Consequently, buyer preference orders for positions are no longer uniform, and hence the separability assumption no longer holds, requiring click-through rate measurement for each position-creative pair. Moreover, it is possible to dynamically re-arrange the content to improve flow (see



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¹The separability assumption says that the click-through rate p_{ij} of ad i in position j can be decomposed into an ad-specific factor α_i and a position specific factor β_j , i.e., $p_{ij} = \alpha_i \cdot \beta_j$ for all ad i and position j .

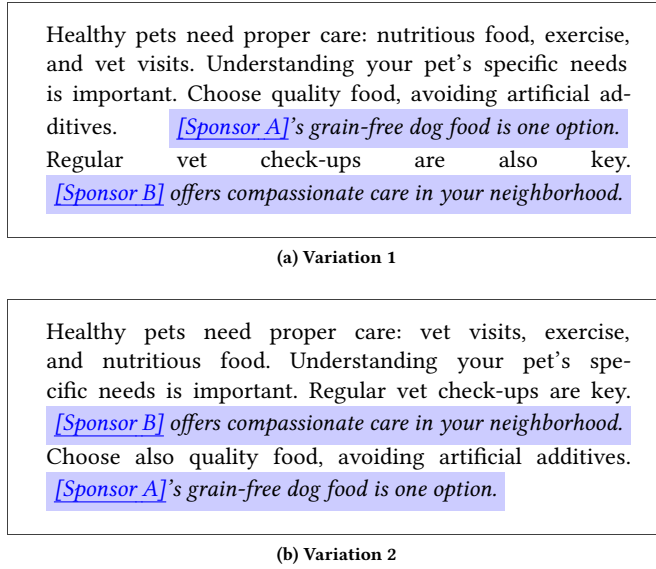


Figure 1: Two variations of ads displayed within AI-generated content.

Figure 1). Fortunately, the advanced semantic understanding of large language models makes this feasible.

Dispensing with the separability assumption introduces two significant challenges. First, the underlying optimization shifts from a straightforward ranking problem to a general matching problem, significantly increasing computational complexity. Second, this formulation necessitates an explicit model of user behavior to rigorously quantify substitution effects, capturing how the interplay between multiple sponsored creatives and the AI-generated context influences user choice.

In this study, we address the design of optimal welfare-maximization and revenue-maximization position auctions within the context of two user behavior models: (1) the canonical *multinomial logit (MNL) model* [28, 30, 33] and (2) the *cascade model* [13]. The MNL model implies a “scan-and-pick” behavior where the user treats the generated content as a single consideration set, evaluating all options simultaneously before making a choice. The cascade model captures the linear consumption nature of text-based AI responses, assuming users read sequentially from top to bottom and stop processing content as soon as a relevant link is clicked. In the MNL model, the user is insensitive to the presentation order of creatives, while in the cascade model the order matters.

In particular, computational tractability is a primary concern, which ensures the practical viability of the proposed auction mechanisms. Therefore, we focus on the design of computationally efficient algorithms, which can be used to obtain optimal welfare-maximization and revenue-maximization position auctions via Vickrey-Clarke-Groves (VCG) mechanisms [12, 41] and Myerson auctions [32].

1.1 Our Contribution

Model Formalization. In Section 2, we formalize the model of the extended position auctions, motivated by the novel application

of placing sponsored creatives within AI-generated commercial content. As we drop the separability assumption, the input of the problem naturally includes a matrix of click-through rates and a vector of bids. For each position-creative pair, the click-through rate represents the probability of the creative being clicked when shown in the corresponding position while no other creative is shown, which we call the *standalone click-through rate*.

In the case where multiple creatives are shown, the user’s attention is distributed over them, and hence the substitution effect emerges. To quantify such effects, certain *user behavior models* must be introduced, and in particular, we study the multinomial logit (MNL) model and the cascade model, respectively. Given a user behavior model, the final click-through rates for each creative is then a function of both the allocation of creatives and their standalone click-through rates in the allocated positions.

Focusing on the ad placement and content arrangement stage, rather than the token-by-token execution stage, offers significant advantages. First, it allows us to abstract away the complexities of token generation, effectively treating the generative model as a black box. Second, this approach decouples the economic mechanism (the auction) from the latency-sensitive inference process, enabling complex welfare calculations without stalling the content stream.

Winner Determination Problem. In Section 3.1, we formalize the *winner determination problem* (WDP), prove the monotonicity of its solution, and demonstrate the reduction from the welfare-maximization and the revenue-maximization problem to it.

The goal of the WDP is to find the “winning” allocation out of the feasible set of allocations that maximizes the total payoff assuming winning advertisers pay their bids for any given “bid” vector. With an oracle that can solve the WDP exactly, one can easily implement the VCG mechanism: the allocation is the solution of the WDP with advertiser values as the “bids”, and the VCG payment can be computed from the optimal payoff of the leave-one-out WDPs.

For the revenue-maximization problem, solving the WDP with advertiser *virtual values* as the “bids” yields the optimal allocation, and the corresponding payment can be computed following the envelope theorem [31]. According to the envelope theorem, one must prove that the allocation function is monotonic in the sense that each advertiser’s click-through rate (weakly) increases as its bid increases (while other bids fixed), and then the corresponding payment can be computed based on the integral of the allocation function. Fortunately, the monotonicity is automatically guaranteed as long as the WDP is solved exactly (Lemma 3.1).

A useful side note is that the envelope theorem works even when the WDP is solved approximately, except that the monotonicity must be proved separately for the implied allocation function. Thus, when an approximate solver for the WDP is given, one can invoke the envelope theorem to construct approximately welfare-maximization and revenue-maximization mechanism as long as the allocation function is monotonic.

Optimal Auctions in MNL Model. In Section 4, we present the welfare- and revenue-maximization mechanisms under the MNL model (Theorem 4.1). Given the reduction to the WDP described above, it remains to provide a computationally efficient algorithm that can solve the WDP exactly. The WDP itself is a mixed integer

program with a non-linear objective. Simple relaxations do not make it a linear program. Leveraging the order-insensitivity of the MNL model, we provide a relaxed linear programming formulation that allows to construct an optimal integral solution to the WDP from any solution of the relaxed linear program (Lemma 4.2).

Approximately Optimal Auctions in Cascade Model. In Section 5, we present the approximately optimal welfare-maximization mechanism and the revenue-maximization mechanism under the cascade model (Corollary 5.1). Similar to the mechanism design under the MNL model, the key is an efficient algorithm that solves the WDP approximately. The cascade model is more challenging because the permutation of the content and the order in which creatives are displayed impacts user choice. To achieve this, we introduce an intermediate problem that 4-approximates the WDP and then design algorithms to solve the intermediate problem (see Section 5.2). Interestingly, the intermediate problem is similar to the so-called *budgeted matching problem* [6], and we present a PTAS that solves it up to any $\epsilon > 0$ gap. However, the resulting allocation function is not necessarily monotonic. As an alternative, we propose an efficient algorithm with an approximation ratio of $O(\ln m)$, where m is the number of positions, and prove the monotonicity of the corresponding allocation rule. We leave it as an interesting open problem to close the gap between the WDP and the WDP with a monotone allocation constraint, particularly, whether it is possible to obtain a monotonic allocation with a constant approximation ratio.

1.2 Related Work

Position Auctions and Online Advertising. In seminal work, Varian [40] first introduced the game-theoretic model of position auctions for sponsored search. Edelman et al. [17] provided a comprehensive analysis of the Generalized Second-Price auction, examining its efficiency and the strategic behavior of advertisers. Blumrosen et al. [7] introduced a setting with non-uniform conversion rates, where the conversion probability given a click varies depending on the ad’s position. Athey and Ellison [3] further advanced the field by incorporating consumer search behavior into the model, recognizing that consumer choices and search strategies influence the value of different ad positions. Thompson and Leyton-Brown [39] developed “computational mechanism analysis” to quantitatively compare different position auction formats, providing insights into their efficiency and effectiveness. More recently, Gravin et al. [22] addressed the bidder selection problem, proposing efficient algorithms for selecting a subset of advertisers to participate in the auction.

User Behavior Model. The cascade model [13], in which users scan slots sequentially from top to bottom, has been shown empirically to explain user choices effectively. Several studies have explored the cascade model and its extensions (see, e.g., [2, 11, 21, 25]). Kempe and Mahdian [25] and Aggarwal et al. [2] demonstrate that the winner determination problem for the cascade model can be solved in polynomial time using dynamic programming. Interestingly, Kempe and Mahdian [25] also examines a more general model with position multipliers, presenting a 4-approximation algorithm and a

QPTAS, but leaves open the question of whether the problem is NP-hard.

In contrast, non-sequential choice models propose that consumers evaluate all advertisements before making a selection. Jeziorski and Segal [24] provides empirical evidence supporting that non-sequential models are better at explaining consumer choices. Ghosh and Mahdian [20] investigates a non-sequential model with a public choice model and private advertiser values, showing that an efficient allocation can be implemented with a VCG mechanism. While the winner determination problem can be NP-hard in general, Ghosh and Mahdian [20] offers a polynomial time algorithm using dynamic programming when preferences are single-peaked. In the previous results, the choice probabilities are public, and the only private information is the advertisers’ values for being chosen. Athey and Ellison [3] give a notable exception, studying the bidding equilibrium under a cascade model where consumers need to click on ads to discover their quality. Balseiro and Désir [5] studies a mechanism design problem for sponsored products in which advertisers hold private information about how their own products match users’ interests.

LLM-motivated mechanism design. Inspired by the increasing adoption of large language models, many recent works look into economic problems that intersect with generative AI. Dubey et al. [15], Duetting et al. [16], Hajiaghayi et al. [23], Soumalias et al. [37] consider mechanism design in the presence of LLMs as an extension for online advertising with generative AI. Sun et al. [38] study the mechanism design problem with strategic agents during the LLM fine-tuning stage. Lu et al. [27], Wu and Hartline [42] leverage the semantic understanding ability of LLMs to expand informative elicitation mechanisms to natural languages. Besides the above mechanism design problems, there are works focusing on understanding the economic behaviors and strategic abilities of LLMs, e.g., [4, 10, 14, 18, 19, 26, 29, 34].

2 Model

We introduce a model of extended position auctions with m positions and n advertisers, where the rendering order of positions is chosen by the auctioneer. In the application of position auctions within AI-generated commercial content, a position consists of the nearby context and a potential slot for sponsored creatives. In particular, the auctioneer can choose an order of the contents (and hence the positions) and instruct LLMs to generate the final content aligned with the selected order.

We denote by $\mathcal{X}$ the set of feasible *allocations* of advertisers to position, and Σ the set of feasible rendering orders of positions or simply *permutations*; see Figure 1 for an example of a permutation. We call the combination of an allocation and a permutation an *augmented allocation*. Allocations are feasible if each advertiser is allocated at at most one position, each position is allocated at at most one advertiser, and the total number of allocated positions is at most $K \leq m$. The set of feasible allocations is thus given by $\mathbf{x} \in \{0, 1\}^{n \times m}$ subject to $\sum_{j=1}^m x_{ij} \leq 1 \forall i \in [n]$, $\sum_{i=1}^n x_{ij} \leq 1 \forall j \in [m]$, and $\sum_{i=1}^n \sum_{j=1}^m x_{ij} \leq K$. The set of feasible permutations Σ is simply the set of all permutations over $[m]$.

Advertisers’ private information is their value for a click. We denote by $v_i \geq 0$ the value of advertiser $i \in [n]$. We assume that

values are independently distributed with an absolutely continuous distribution. We denote by F_i the cumulative distribution and by f_i the density of advertiser i 's value, respectively.

Our model allows for externalities between advertisers' click-through rates. We denote by $\pi_i : \mathcal{X} \times \Sigma \rightarrow [0, 1]$ a function that maps the augmented allocation $\mathbf{a} := (\mathbf{x}, \sigma) \in \mathcal{X} \times \Sigma$ to a click-through rate $\pi_i(\mathbf{a})$ or equivalently $\pi_i(\mathbf{x}, \sigma)$. Click-through rates are usually estimated by platforms using sophisticated predictions models and we assume that $\pi_i(\mathbf{x}, \sigma)$ is common knowledge.

By the revelation principle, we can restrict attention to direct mechanisms in which every advertiser reports their valuation. A mechanism is a pair $(\mathbf{q}, \mathbf{t})$ characterized by an augmented allocation function $\mathbf{q} : \mathbb{R}_+^n \rightarrow \Delta(\mathcal{X} \times \Sigma)$ and payment functions $\mathbf{t} : \mathbb{R}_+^n \rightarrow \mathbb{R}_+^n$. The augmented allocation function maps a vector of reported values $\mathbf{v} \in \mathbb{R}_+^n$ to a lottery over augmented allocations $\mathbf{q}(\mathbf{v}) \in \mathcal{X} \times \Sigma$. We denote by $q(\mathbf{v})[\mathbf{a}]$ or equivalently $q(\mathbf{v})[\mathbf{x}, \sigma]$ the probability of choosing augmented allocation $(\mathbf{x}, \sigma) \in \mathcal{X} \times \Sigma$. The payment function gives the expected payment $t_i(\mathbf{v})$ made by advertiser i when reported values are $\mathbf{v} \in \mathbb{R}_+^n$.

For simplicity, we may abuse the above notations by omitting σ (e.g., $\pi_i(\mathbf{x}, \sigma)$ as $\pi_i(\mathbf{x})$, $q(\mathbf{v})[\mathbf{x}, \sigma]$ as $q(\mathbf{v})[\mathbf{x}]$, $\mathcal{X} \times \Sigma$ as $\mathcal{X}$) when the value of σ is clear from the context or the outcomes are insensitive to σ , e.g., $\pi_i(\mathbf{x}, \sigma) \equiv \pi_i(\mathbf{x})$, $\forall \sigma \in \Sigma$.

Advertisers are quasi-linear utility maximizers, i.e., they maximize the difference between their expected value and the payments made to the mechanism. The expected utility of advertiser i when their value is $v_i \in \mathbb{R}_+$ and the vector of reported values of $\hat{\mathbf{v}} \in \mathbb{R}_+^n$ is

$$u_i(\hat{\mathbf{v}}, v_i) = v_i \sum_{\mathbf{x} \in \mathcal{X}, \sigma \in \Sigma} q(\hat{\mathbf{v}})[\mathbf{x}, \sigma] \pi_i(\mathbf{x}, \sigma) - t_i(\hat{\mathbf{v}}).$$

We restrict attention to mechanisms that are dominant-strategy incentive compatible (IC) and individually rational (IR). A mechanism is IC if every buyer is better off reporting their value truthfully regardless of the reported values of their competitors. That is,

$$u_i((v_i, \hat{\mathbf{v}}_{-i}), v_i) \geq u_i(\hat{\mathbf{v}}, v_i), \quad \forall i \in [n], \hat{\mathbf{v}} \in \mathbb{R}_+^n, v \in \mathbb{R}_+, \quad (\text{IC})$$

where we denote by $\hat{\mathbf{v}}_{-i} \in \mathbb{R}_+^{n-1}$ the vector of advertisers values excluding advertiser i . A mechanism is IR if advertisers get a non-negative utility from participating

$$u_i((v_i, \hat{\mathbf{v}}_{-i}), v_i) \geq 0, \quad \forall i \in [n], \hat{\mathbf{v}} \in \mathbb{R}_+^n, v \in \mathbb{R}_+. \quad (\text{IR})$$

2.1 User Behavior Models

We consider different models for click-through rate that incorporate user behavior. The models take as input a matrix $\mathbf{p} \in \mathbb{R}^{n \times m}$, where p_{ij} is the probability that a user clicks on advertisement i when displayed in position j when no other advertisement is shown in any other position. A key advantage of utilizing $\mathbf{p}$ as input is that it can be accurately estimated by existing state-of-the-art models. We consider two popular user behavior model that incorporate substitution effects between advertisers: a multinomial logit model and a cascade model.

Multinomial Logit Model (MNL). The MNL model [28, 30, 33] assumes that the user obtains independent random utilities from each ad, which are drawn from independent Gumbel distributions. The user considers all advertisements simultaneously and chooses one with the highest utility. MNL models are used extensively in

marketing, economics, and operations literature. These models are popular because they are interpretable, flexible, easy to estimate, and extendable. Let $\rho_{ij} = \text{logit}(p_{ij}) = \log(p_{ij}/(1 - p_{ij}))$ be the log-odds of showing advertiser i in slot j . The click-through rate of advertiser i under an augmented allocation $(\mathbf{x}, \sigma)$ is

$$\pi_i(\mathbf{x}, \sigma) \equiv \pi_i(\mathbf{x}) = \frac{\sum_{j \in [m]} x_{ij} \exp(\rho_{ij})}{1 + \sum_{i' \in [n]} \sum_{j \in [m]} x_{i'j} \exp(\rho_{i'j})}.$$

Since the click-through rate here is insensitive to the permutation σ , we will omit σ from now on whenever the MNL model is adopted. When advertiser i is shown in slot j and no ads are shown in other slots, we obtain that $\pi_i(\mathbf{e}^{ij}) = p_{ij}$ where $\mathbf{e}^{ij} \in \mathcal{X}$ is the allocation with one in position (i, j) and zero otherwise.

In the context of AI-generated content, the MNL model implies a “scan-and-pick” consumer behavior. It assumes the user treats the generated content (or the set of embedded creatives) as a single consideration set, evaluating all options simultaneously before making a choice. This behavior is particularly relevant when the AI generates a list of recommendations or when the layout allows the user to glance at multiple embedded links at once. Crucially, this model implies that while the context of a position matters (affecting the base utility), the ordering does not prevent a user from seeing downstream ads; the user's attention is assumed to cover the full set of visible options.

Cascade Model. The cascade model [13] assumes that the user reads through the ads in the given rendering order and leaves the system once a click happens. Consequently, ads placed earlier in the text act as gatekeepers; if a top-positioned creative is highly relevant, it drastically reduces the probability that users will ever observe or consider creatives embedded later in the generated response. This model best fits conversational agents where the user flow is strictly linear.

Assuming all ad-clicking decisions are independent, the click-through rate of advertiser i under allocation $\mathbf{x} \in \mathcal{X}$ and permutation σ is discounted by the product of $(1 - p_{i'j'})$ for all ads i' shown before it, i.e.,

$$\pi_i(\mathbf{x}, \sigma) = \sum_{j \in [m]} x_{ij} p_{ij} \prod_{j': \sigma(j') < \sigma(j)} \left(1 - \sum_{i' \in [n]} x_{i'j'} p_{i'j'} \right).$$

We again have $\pi_i(\mathbf{e}^{ij}, \sigma) = p_{ij}$.

2.2 Objectives

We consider two objectives for the ad platform, welfare and revenue maximization. When the mechanism is IC and IR, the advertisers report truthfully and we can evaluate the metrics by taking expectation over the distribution of advertisers' values.

The welfare is the sum of the platform revenue and the advertiser utility. Since payments are internal transfers of wealth, they cancel out and the expected welfare of a mechanism $(\mathbf{q}, \mathbf{t})$ is

$$\text{Welfare}(\mathbf{q}, \mathbf{t}) = \mathbb{E}_{\mathbf{v}} \left[\sum_{i \in [n]} v_i \sum_{\mathbf{x} \in \mathcal{X}, \sigma \in \Sigma} q(\mathbf{v})[\mathbf{x}, \sigma] \pi_i(\mathbf{x}, \sigma) \right].$$

The revenue of the ad platform is just the sum of the advertisers' payments:

$$\text{Revenue}(\mathbf{q}, \mathbf{t}) = \mathbb{E}_{\mathbf{v}} \left[\sum_{i \in [n]} t_i(\mathbf{v}) \right].$$

3 Winner Determination Problem and Reduction from Mechanism Design Problems

In this section, we first formally define the *winner determination problem* (WDP) and then show how the optimal welfare-maximization auction and the optimal revenue-maximization auction are constructed with an exact solver of the WDP. We also extend the reduction to the case where only an approximate solver is given, in which the corresponding auctions downgrade to approximately optimal auctions.

3.1 Winner Determination Problem

The WDP determines, given a vector of “bids” $\mathbf{b} \in \mathbb{R}^n$, the augmented allocation $\mathbf{a} = (\mathbf{x}, \sigma) \in \mathcal{X} \times \Sigma$ that maximizes the payoff assuming advertisers pay their bids. This is given by

$$\mathbf{a}^*(\mathbf{b}) = (\mathbf{x}^*(\mathbf{b}), \sigma^*(\mathbf{b})) \in \arg \max_{\mathbf{a} \in \mathcal{X} \times \Sigma} \sum_{i \in [n]} b_i \pi_i(\mathbf{a}).$$

In following sections, we discuss how to efficiently solve the WDP. Before that, we prove monotonicity of click-through rates of an advertiser in terms of its bid when the WDP is solved exactly. We postpone all proofs to the appendix.

Lemma 3.1. *Fix an advertiser $i \in [n]$. Let $\mathbf{a}^*(\mathbf{b})$ be an optimal augmented allocation for a bid vector $\mathbf{b}$ and $\mathbf{a}^*(\hat{\mathbf{b}})$ be an optimal augmented allocation for the vector $\hat{\mathbf{b}}$ with $\hat{b}_i \geq b_i$ and $\hat{b}_{i'} = b_{i'}$ for $i' \neq i$. Then, $\pi_i(\mathbf{a}^*(\hat{\mathbf{b}})) \geq \pi_i(\mathbf{a}^*(\mathbf{b}))$.*

3.2 Mechanism Design Problem

In this subsection, we show how the welfare-maximization auction and the revenue-maximization auction can be implemented when an exact optimization algorithm of the WDP is given.

Welfare Maximization. The ad platform's problem is to solve

$$\max_{(\mathbf{q}, \mathbf{t}) : (\text{IC}) \text{ and } (\text{IR})} \text{Welfare}(\mathbf{q}, \mathbf{t}).$$

If we ignore the individual rationality (IR) and incentive compatibility (IC) constraints, we can write the problem as

$$\begin{aligned} \max_{(\mathbf{q}, \mathbf{t})} \text{Welfare}(\mathbf{q}, \mathbf{t}) &= \max_{\mathbf{q}} \mathbb{E}_{\mathbf{v}} \left[\sum_{i \in [n]} v_i \sum_{\mathbf{a} \in \mathcal{X} \times \Sigma} q(\mathbf{v})[\mathbf{a}] \pi_i(\mathbf{a}) \right] \\ &= \mathbb{E}_{\mathbf{v}} \left[\max_{\mathbf{a} \in \mathcal{X} \times \Sigma} \sum_{i \in [n]} v_i \pi_i(\mathbf{a}) \right], \end{aligned}$$

where the first equation follows because payments do not enter the objective, and the second because the platform should optimize pointwise over values and pick the allocation with the highest values. Denote the allocation that maximizes welfare for each realization of advertisers' values by $\mathbf{a}^*(\mathbf{v})$.

We can implement a welfare-maximizing augmented allocation $\mathbf{a}^*(\mathbf{v})$ truthfully using the Vickrey–Clarke–Groves (VCG) mechanism. Under the VCG mechanism, the platform asks advertisers to report their values $\mathbf{v}$, and then it chooses the augmented allocation $\mathbf{a}^*(\mathbf{v})$, and charges each advertiser the externality they impose on others. That is, the augmented allocation is $q(\mathbf{v})[\mathbf{a}] = 1$ for $\mathbf{a} = \mathbf{a}^*(\mathbf{v})$ and $q(\mathbf{v})[\mathbf{a}] = 0$ otherwise and the payment of advertiser i is

$$t_i(\mathbf{v}) = \sum_{i' \neq i} v_{i'} \pi_{i'}(\mathbf{a}^*(0, \mathbf{v}_{-i})) - \sum_{i' \neq i} v_{i'} \pi_{i'}(\mathbf{a}^*(\mathbf{v})). \quad (1)$$

The VCG mechanism is IC. Because values are non-negative, we have that payments satisfy $t_i(\mathbf{v}) \geq 0$ and the mechanism is also IR.

Revenue Maximization. The ad platform problem is to solve

$$\max_{(\mathbf{q}, \mathbf{t}) : (\text{IC}) \text{ and } (\text{IR})} \text{Revenue}(\mathbf{q}, \mathbf{t}).$$

We can use the Envelope theorem [31] to characterize optimal mechanisms. Let

$$y_i(\mathbf{v}) = \sum_{\mathbf{a} \in \mathcal{X} \times \Sigma} q(\mathbf{v})[\mathbf{a}] \pi_i(\mathbf{a}),$$

be the probability that advertiser i is allocated (in any slot) under a mechanism $(\mathbf{q}, \mathbf{t})$. We refer to $y_i(\mathbf{v})$ as the *total click probability* of advertiser i . Then, a mechanism is IC if and only if the total click probability $y_i(\mathbf{v})$ is non-decreasing in v_i for all advertiser $i \in [n]$ and payments satisfy

$$t_i(\mathbf{v}) = t_i(0, \mathbf{v}_{-i}) + v_i y_i(\mathbf{v}) - \int_0^{v_i} y_i(z, \mathbf{v}_{-i}) dz. \quad (2)$$

Moreover, a mechanism is IR if $t_i(0, \mathbf{v}_{-i}) \leq 0$. At optimality, we should set the payment of the lowest type to zero, i.e., $t_i(0, \mathbf{v}_{-i}) = 0$. Using integration by parts, we can write the revenue of an optimal mechanism as

$$\begin{aligned} \text{Revenue}(\mathbf{q}, \mathbf{t}) &= \mathbb{E}_{\mathbf{v}} \left[\sum_{i \in [n]} \left(v_i - \frac{1 - F_i(v_i)}{f_i(v_i)} \right) y_i(\mathbf{v}) \right] \\ &= \mathbb{E}_{\mathbf{v}} \left[\sum_{i \in [n]} \phi_i(v_i) \sum_{\mathbf{a} \in \mathcal{X} \times \Sigma} q(\mathbf{v})[\mathbf{a}] \pi_i(\mathbf{a}) \right], \end{aligned}$$

where the last equation follows from using our formula for the total click probability $y_i(\mathbf{v})$ and letting $\phi_i(v_i) = v_i - (1 - F_i(v_i))/f_i(v_i)$ be the virtual value of advertiser $i \in [n]$.

A similar argument to the welfare case implies that the platform should choose the allocation that maximizes virtual values, i.e., choose $\mathbf{a}^*(\mathbf{b})$ with $b_i = \phi_i(v_i)$. This allocation can be implemented truthfully if payments are chosen to satisfy (2) and the total click probability is non-decreasing. A sufficient condition for the total click probability $y_i(\mathbf{v})$ to be monotonic is that virtual values $\phi_i(v_i)$ are non-decreasing and that $\pi_i(\mathbf{a}^*(\mathbf{v}))$ is non-decreasing in v_i for every advertiser i and valuations $\mathbf{v}_{-i}$ of competitors.

3.3 Mechanism Design with Approximate WDP Solutions

A useful observation is that the envelope theorem works even when the WDP is solved approximately. However, additional care

is needed for proving the monotonicity of the induced allocation function, as Lemma 3.1 does not apply to approximate solvers.

Following the reductions in Section 3.2, an approximately optimal solution to the WDP yields approximately optimal welfare (or revenue) from the mechanism, and the corresponding payment is computed according to the envelope theorem.

4 Multinomial Logit Model

A special case of the MNL model is when the log-odds ρ_{ij} are additively separable, i.e., $\rho_{ij} = \rho_i + \rho_j$. Abeliuk et al. [1] showed that the winner determination problem can be solved in quadratic time when log-odds are additively separable. Additive separability allows no interaction between positions and advertisers, as every advertiser equally prefers each position. We next discuss how to solve the WDP for general log-odds efficiently. Since the MNL model is invariant to permutations, we omit the permutation σ and focus on the allocation x for the entire section.

The key of the results in this section is an efficient algorithm solving the WDP for general log-odds. Taking the algorithm as given, with the reduction introduced in Section 3.2, we can implement both the welfare-maximization auction and the revenue-maximization auction. Formally, we have the following theorem:

THEOREM 4.1 (OPTIMAL AUCTIONS IN MNL MODEL). *Under the MNL model, there exists:*

- A welfare-maximization auction that is computationally efficient and satisfies IC and IR;
- A revenue-maximization auction that is computationally efficient and satisfies ϵ -IC and IR.

The ϵ loss on IC for the revenue-maximization auction comes from the ϵ -accuracy loss of payment computation, where one needs to compute the integral of the allocation function with potentially exponentially many pieces. The ϵ loss then comes from discretizing the space of the integral to guarantee computational efficiency. We now proceed to discuss how to efficiently solve the WDP under the MNL model, in spite of the non-linear objective:

$$\max_{x \in \mathcal{X}} \frac{\sum_{i \in [n]} \sum_{j \in [m]} b_i x_{ij} \exp(\rho_{ij})}{1 + \sum_{i' \in [n]} \sum_{j \in [m]} x_{i'j} \exp(\rho_{i'j})}.$$

Consider the relaxation in which we optimize over the convex hull of $\mathcal{X}$ given by $\tilde{\mathcal{X}} = \{x \in \mathbb{R}_+^{n \times m}\}$ subject to $\sum_{j=1}^m x_{ij} \leq 1 \forall i \in [n]$, $\sum_{i=1}^n x_{ij} \leq 1 \forall j \in [m]$, and $\sum_{i=1}^n \sum_{j=1}^m x_{ij} \leq K$. We present a Charnes-Cooper transformation [9] that allows us to write the linear-fractional problem as a linear program. introduce the following variables

$$y_{ij} = \frac{x_{ij}}{1 + \sum_{i' \in [n]} \sum_{j \in [m]} x_{i'j} \exp(\rho_{i'j})},$$

$$z = \frac{1}{1 + \sum_{i' \in [n]} \sum_{j \in [m]} x_{i'j} \exp(\rho_{i'j})}.$$

We can equivalently solve the following linear program:

$$\begin{aligned} & \max_{y \in \mathbb{R}_+^{n \times m}, z \in \mathbb{R}} \sum_{i \in [n]} \sum_{j \in [m]} b_i y_{ij} \exp(\rho_{ij}) \\ & \text{s.t.} \sum_{j=1}^m y_{ij} \leq z, \forall i \in [n], \\ & \sum_{i=1}^n y_{ij} \leq z, \forall j \in [m], \\ & \sum_{i=1}^n \sum_{j=1}^m y_{ij} \leq Kz, \\ & \sum_{i \in [n]} \sum_{j \in [m]} y_{ij} \exp(\rho_{ij}) + z = 1. \end{aligned} \quad (3)$$

Our approach requires an extreme point solution to (3), which can be obtained by the simplex method, or using an interior point method together with a basis recovery approach.

Lemma 4.2. *Suppose $b \neq 0$. Let (y, z) be an extreme point optimal solution to (3). Then, an optimal integral matching can be obtained by setting $x = \frac{y}{z}$.*

5 Cascade Model

In this section, we consider the Cascade model. In particular, we focus on the design of efficient approximate algorithms for the WDP. In light of the reduction from Section 3.3, it suffices to show that the induced allocation function is monotonic and that it approximates the optimal objective value.

For ease of notation, we take the values v as the “bids” of the WDP and hence the optimization target becomes the social welfare $\text{Welfare}(x, \sigma) = \sum_i v_i \cdot \pi_i(x, \sigma)$. Without loss of generality, we assume $v_1 \geq v_2 \geq \dots \geq v_n$.

The rest of the section is organized as follows. In Section 5.1, we simplify the objective by fixing the optimal permutation. In Section 5.2, we introduce an intermediate problem $\text{Welfare}^{\text{restricted}}$ which 4-approximates our original WDP problem, and it is simpler in the sense that it avoids the cascading definition. In Section 5.3, we give a PTAS for $\text{Welfare}^{\text{restricted}}$ (Theorem 5.12) and this implies a constant approximation algorithm for WDP (Corollary 5.13). Unfortunately, we cannot prove that it gives a monotonic allocation. Finally, in Section 5.4, we give an approximation algorithm to WDP with a monotonic allocation (Theorem 5.14), with a weaker approximation ratio $O(\ln m)$. As a corollary, we obtain the following.

Corollary 5.1. *Under the Cascade model, we have an $O(\ln m)$ -approximate, computationally efficient, ϵ -IC, and IR auction for welfare maximization or revenue maximization, respectively.*

We leave it as an interesting open problem to close the gap between the approximation ratio of the WDP and the approximation ratio of the WDP with monotonic allocation.

5.1 Optimal Permutation

We first pin down the choice of the optimal permutation, which turns out to be agnostic to the implementation details of the allocation function. Therefore, we can focus on the allocation design and simplify the notations.

First, when a slot j is not allocated (that is, $\sum_{i \in [n]} x_{ij} = 0$), shuffling the slot up or down in the permutation does not change the welfare. Without loss of generality, we only consider permutations which put unallocated slots in the end, and call this set of permutations Σ^* . Next, for these permutations, we show a lemma about the permutation maximizing welfare.

Lemma 5.2. *For a fixed $\mathbf{x} \in \mathcal{X}$, any permutation $\sigma^* \in \Sigma^*$ maximizing $\text{Welfare}(\mathbf{x}, \sigma)$ sorts slots in decreasing order of their matched values, i.e., for any j and j' with $\sum_{i \in [n]} v_i x_{ij} > \sum_{i \in [n]} v_i x_{ij'}$, $\sigma^*(j) < \sigma^*(j')$.*

In the rest of the section, we will only consider the welfare-maximizing permutation which sorts its matched value in decreasing order (without loss of generality, we break ties of equal matched values by advertiser indices). We omit σ in notation π and the corresponding Welfare. We have

$$\pi_i(\mathbf{x}) = \sum_{j \in [m]} x_{ij} \cdot p_{ij} \cdot \prod_{i' < i} \left(1 - \sum_{j' \in [m]} x_{i'j'} p_{i'j'} \right),$$

and this implies

$$\pi_i(\mathbf{x}) = \sum_{j \in [m]} x_{ij} \cdot p_{ij} \cdot \left(1 - \sum_{i' < i} \pi_{i'}(\mathbf{x}) \right).$$

5.2 A 4-approximation via Restricted Welfare

In this section, we introduce a restricted welfare problem $\text{Welfare}^{\text{restricted}}$ and show it gives a 4-approximation to the WDP in the Cascade model.

The restricted welfare is defined as follows. We define an auxiliary clickthrough rate $\pi^{\text{restricted}}$ that does not cascade and only counts welfare for top slots up to total click-through rate 1, in contrast to the click-through rate π . More formally,

$$\pi_i^{\text{restricted}}(\mathbf{x}) = \sum_{j \in [m]} x_{ij} \cdot \min \left(p_{ij}, 1 - \sum_{i' < i} \pi_{i'}^{\text{restricted}}(\mathbf{x}) \right).$$

We define the corresponding $\text{Welfare}^{\text{restricted}}(\mathbf{x})$ as $\text{Welfare}(\mathbf{x})$ with π replaced by $\pi^{\text{restricted}}$. We first show a simple lemma to compare $\pi(\mathbf{x})$ and $\pi^{\text{restricted}}(\mathbf{x})$:

Lemma 5.3. *For every $n' \in [n]$, we have $\sum_{i=1}^{n'} \pi_i(\mathbf{x}) \leq \sum_{i=1}^{n'} \pi_i^{\text{restricted}}(\mathbf{x}) \leq 1$, and $\pi_i^{\text{restricted}}(\mathbf{x}) \geq 0$ for $i \in [n]$.*

We then prove in Lemma 5.5 and Lemma 5.4 that $\text{Welfare}^{\text{restricted}}(\mathbf{x})$ is a 4-approximation of $\text{Welfare}(\mathbf{x})$.

Lemma 5.4. *For any $\mathbf{x} \in \mathcal{X}$,*

$$\text{Welfare}^{\text{restricted}}(\mathbf{x}) \geq \text{Welfare}(\mathbf{x}).$$

Lemma 5.5. *For any $\mathbf{x} \in \mathcal{X}$,*

$$\text{Welfare}^{\text{restricted}}(\mathbf{x})/4 \leq \text{Welfare}(\mathbf{x}).$$

With Lemma 5.5 and Lemma 5.4, we know that if we optimize $\text{Welfare}^{\text{restricted}}(\mathbf{x})$, we get a 4-approximation to $\text{Welfare}(\mathbf{x})$. In the next section, we provide an algorithm to approximately optimize $\text{Welfare}^{\text{restricted}}(\mathbf{x})$.

5.3 A PTAS for $\text{Welfare}^{\text{restricted}}$

We give a PTAS for $\text{Welfare}^{\text{restricted}}$ in this section using an algorithm from [6] for the budgeted matching problem, which is similar to the restricted welfare problem. The budgeted matching problem can be defined as follows: let $\pi_i^{\text{budgeted}}(\mathbf{x}) = \sum_{j \in [m]} x_{ij} p_{ij}$ and define $\text{Welfare}^{\text{budgeted}}(\mathbf{x})$ as $\text{Welfare}(\mathbf{x})$ with π replaced by π^{budgeted} . The goal of the budgeted matching problem is to find an integer solution $\mathbf{x}$ maximizing $\text{Welfare}^{\text{budgeted}}(\mathbf{x})$ subject to $\sum_{i \in [n]} \pi_i^{\text{budgeted}}(\mathbf{x}) \leq 1$.

THEOREM 5.6 (THEOREM 1 OF [6]). *There is a PTAS for the budgeted matching problem.*

Here we give a PTAS algorithm for $\text{Welfare}^{\text{restricted}}$. The main idea of the algorithm is to guess the “discounted advertiser” (as later defined in Definition 5.7) and how much it is discounted, and then apply the PTAS of the budget matching problem. Recall that in the restricted welfare model, user attention is treated as a finite resource capped at a total probability of 1. Intuitively, the discounted advertiser is the single ad that straddles this limit.

ALGORITHM 1: PTAS for $\text{Welfare}^{\text{restricted}}$

Input: Approximation parameter $\varepsilon \in (0, 1)$, number of advertisers n , number of positions m , values v_i for $i \in [n]$, base click-through rates p_{ij} for $i \in [n], j \in [m]$

Output: Allocation $\mathbf{x}$

Initialize $\mathbf{x}$ to be an arbitrary allocation

for $k = 1, \dots, n$ **do**

for $\alpha = \varepsilon/2, \varepsilon, \dots, \lfloor 2/\varepsilon \rfloor \cdot \varepsilon/2, 1$ **do**

 Define $p_{ij}^{k,\alpha} = p_{ij}$ for $i \in [n] \setminus \{k\}, j \in [m]$, and

$p_{ij}^{k,\alpha} = p_{ij} \cdot \alpha$ for $i = k, j \in [m]$

 Solve the budget matching with $p^{k,\alpha}$ as base click-through rates ($\text{Welfare}^{\text{budgeted}-k,\alpha}$) using the PTAS from [6] to get a $(1 - \varepsilon/2)$ -approximation $\mathbf{x}^{k,\alpha}$

if $\text{Welfare}^{\text{restricted}}(\mathbf{x}^{k,\alpha}) \geq \text{Welfare}^{\text{restricted}}(\mathbf{x})$ **then**

$\mathbf{x} \leftarrow \mathbf{x}^{k,\alpha}$

Output $\mathbf{x}$

Before we analyze Algorithm 1, we provide auxiliary definitions and lemmas.

Definition 5.7. *For any $\mathbf{x} \in \mathcal{X}$, call an advertiser k a discounted advertiser if $\pi_k^{\text{restricted}}(\mathbf{x}) > 0$ and $\pi_k^{\text{restricted}}(\mathbf{x}) < \pi_k^{\text{budgeted}}(\mathbf{x}) = \sum_{j \in [m]} x_{kj} p_{kj}$.*

Lemma 5.8. *For any $\mathbf{x} \in \mathcal{X}$, there exists at most one discounted advertiser.*

Definition 5.9. *For any $\mathbf{x} \in \mathcal{X}$, define $\text{ZS}(\mathbf{x})$ to be the procedure of forcing advertiser i 's allocation to be 0 if $\pi_i^{\text{restricted}}(\mathbf{x}) = 0$, i.e.,*

- $\text{ZS}(\mathbf{x})_{ij} = x_{ij}$ if $\pi_i^{\text{restricted}}(\mathbf{x}) > 0$, and
- $\text{ZS}(\mathbf{x})_{ij} = 0$ if $\pi_i^{\text{restricted}}(\mathbf{x}) = 0$.

Lemma 5.10. For any $x \in \mathcal{X}$, $x' = \text{ZS}(x)$ has the following two properties:

- For every advertiser i , we have $\pi_i^{\text{restricted}}(x) = \pi_i^{\text{restricted}}(x')$
 $\text{Welfare}^{\text{restricted}}(x) = \text{Welfare}^{\text{restricted}}(x')$.
- For any advertiser i that is not a discounted advertiser in x ,
 $\pi_i^{\text{restricted}}(x') = \pi_i^{\text{budgeted}}(x')$.

Lemma 5.11. For any $x \in \mathcal{X}$ satisfying $\sum_{i=1}^n \pi_i^{\text{budgeted}-k, \alpha}(x) \leq 1$, we have

$$\text{Welfare}^{\text{budgeted}-k, \alpha}(x) \leq \text{Welfare}^{\text{restricted}}(x).$$

Now we are ready to state the main theorem for Algorithm 1.

THEOREM 5.12. Algorithm 1 is a PTAS for $\text{Welfare}^{\text{restricted}}$, i.e.,

- Algorithm 1 runs in time polynomial in n and m for a fixed ϵ /
- The output of Algorithm 1 has $\text{Welfare}^{\text{restricted}}$ at least $(1 - \epsilon)$ times the optimal $\text{Welfare}^{\text{restricted}}$.

Combining Theorem 5.12 and the results of Section 5.2 (Lemma 5.4 and 5.5), we get the following corollary.

Corollary 5.13. Algorithm 1 gives a $\frac{4}{1-\epsilon}$ -approximation to the optimal Welfare in the Cascade model.

5.4 Monotonic $O(\ln m)$ -approximation of the WDP

In this section, we give a randomized $O(\ln m)$ -approximate algorithm for the WDP, which induces a monotonic allocation function. According to Section 3.3, the corresponding welfare-maximization auction and revenue-maximization auction are implied with the same approximation guarantee.

THEOREM 5.14. There is a polynomial-time randomized algorithm that finds an allocation with expected approximation ratio of $O(\ln m)$ to the optimal welfare in the Cascade model. Moreover, the allocation algorithm is monotonic.

Similar to our approach in Section 5.3, we work with restricted welfare and the budgeted matching problem using the base click-through rate p_{ij} 's instead of the actual (cascade) click-through rate. A naive way to address the budget constraint is to only pick a single edge which must be feasible since $p_{ij} \leq 1$. The edge with maximum $v_i \cdot p_{ij}$ is a trivial m -approximation. Note that in the previous argument, we essentially turn the budget constraint into a cardinality constraint (of 1 edge), and we can build upon this idea to get the stronger $O(\ln m)$ -approximation by bucketizing the edges by their p_{ij} 's.

In particular, we partition the advertiser-position pairs (i.e. edges) into $G_1, \dots, G_{\log_2(4m)}$ (where for notational simplicity we assume m is a power of 2). The subgraph G_ℓ contains all edges with $\frac{1}{2^\ell} < p_{ij} \leq \frac{1}{2^{\ell-1}}$ for $\ell \in [1, \log_2(2m)]$, and the last subgraph $G_{\log_2(4m)}$ contains all remaining edges (i.e., those with $p_{ij} \leq 1/(2m)$).

If we can find the allocation with the optimal welfare in each subgraph, the sum of these allocations' welfare is at least the welfare of the optimal allocation in the original graph (i.e. with all edges), and this gives us the following.

Lemma 5.15. If we can find a β -approximately optimal welfare allocation in each subgraph, then the algorithm of picking a bucket $\ell \in$

$[1, \log_2(4m)]$ uniformly at random and using the β -approximately optimal allocation of G_ℓ would have an expected $\beta \cdot \log_2(4m)$ approximation ratio to the optimal welfare in the full graph. Furthermore, if the algorithm to find the allocation in each subgraph is monotonic, the overall algorithm (i.e., with the random picking) is also monotonic.

This reduces our task to finding a monotonic allocation algorithm that approximately optimizes the welfare in a single subgraph. To make our allocation algorithm monotonic, we need to explicitly specify the permutation σ of the matched positions, which may not be the same as the optimal permutation that ranks the positions in decreasing order of their matched advertisers' value.

We use the (canonical) greedy algorithm to find a *maximal* matching subject to cardinality constraint in the subgraph G_ℓ , and use the same order of edges being added to the matching as the permutation of the matched positions.

ALGORITHM 2: Greedy maximal matching in G_ℓ

Input: Index $\ell \in [1, \log_2(4m)]$, advertiser values v_i for $i \in [n]$, and p_{ij} 's for edges (i, j) in G_ℓ .

Output: Matching $\mathbf{x}_\ell$ and permutation σ_ℓ of matched positions.

Start with an empty matching $\mathbf{x}_\ell$ and number of matched edges $k = 0$
 Go through the edges in decreasing order of weights $w_{ij} = v_i \cdot p_{ij}$
 (with any deterministic tie-breaking)

while There are still edges to consider, denote (i, j) as the next edge **do**
 if (i, j) can be added to $\mathbf{x}_\ell$ (i.e. both advertiser i and position j are unmatched) **then**
 $k \leftarrow k + 1$, $\sigma_\ell(j) \leftarrow k$
 Add the edge (i, j) to $\mathbf{x}_\ell$
 if $k = \min(2^\ell, m)$ **then**
 Return $\mathbf{x}_\ell$ and σ_ℓ

Return $\mathbf{x}_\ell$ and σ_ℓ

Lemma 5.16. For any subgraph G_ℓ , Algorithm 2 (in polynomial time) finds an allocation of welfare at least $\frac{1}{28}$ of the optimal allocation in G_ℓ , and the allocation algorithm is monotonic.

It is straightforward to see that Theorem 5.14 follows from Lemma 5.15 and Lemma 5.16.

6 Conclusions

In this work, we formalized the problem of position auctions within AI-generated content by relaxing the standard separability assumption in position auctions to capture substitution effects between advertisers. We designed computationally efficient, incentive-compatible mechanisms that achieve optimal welfare and revenue for the MNL and Cascade models.

Several promising directions remain for future research. First, regarding the technical results, a significant open problem is closing the gap between the constant-factor approximation of the winner determination problem and the $O(\ln m)$ approximation currently required to ensure a monotonic allocation rule under the Cascade model. Second, regarding user behavior, future work could extend this framework to richer consumer search models, such as incorporating rational search behavior where users update their beliefs and stopping rules dynamically based on content quality. Addressing these complexities would further bridge the gap between economic search theory and the nascent field of advertising in generative AI.

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A Missing proofs of Section 3.1 and Section 4

PROOF OF LEMMA 3.1. Let $\pi = (\pi_i(a^*(b)))_{i \in [n]}$ and $\hat{\pi} = (\pi_i(a^*(\hat{b})))_{i \in [n]}$. We prove the result by contradiction. Suppose $\hat{b}_i > b_i$ but $\hat{\pi}_i < \pi_i$. By the optimality of $a^*(\hat{b})$, we have that $\hat{b}_i \hat{\pi}_i + \sum_{i' \neq i} b_{i'} \hat{\pi}_{i'} \geq \hat{b}_i \pi_i + \sum_{i' \neq i} b_{i'} \pi_{i'}$. Because $\hat{b}_i > b_i$ and $\hat{\pi}_i < \pi_i$, we have $(\hat{b}_i - b_i) \cdot (\hat{\pi}_i - \pi_i) < 0$, which implies after re-arranging that $b_i \hat{\pi}_i + \hat{b}_i \pi_i > \hat{b}_i \hat{\pi}_i + b_i \pi_i$. Summing together, we obtain that $b_i \hat{\pi}_i + \sum_{i' \neq i} b_{i'} \hat{\pi}_{i'} > b_i \pi_i + \sum_{i' \neq i} b_{i'} \pi_{i'}$, which contradicts the optimality of $a^*(b)$. $\square$

PROOF OF LEMMA 4.2. We first argue that every extremal optimal solution of the relaxed WDP $\max_{x \in \bar{X}} \sum_{i \in [n]} b_i \pi_i(x)$ is optimal for the original WDP $\max_{x \in X} \sum_{i \in [n]} b_i \pi_i(x)$. Because the extreme points of the matching polytope $\bar{X}$ are integral matchings, hence an extreme point $x \in \bar{X}$ is feasible for the WDP, i.e., $x \in X$. The claim then follows because both problems have the same objective function and $X \subseteq \bar{X}$.

Let (y, z) be an extremal optimal solution to (3). We cannot have $z < 0$ because $y \geq 0$. If we have $z = 0$, then $y = 0$, which can never be optimal because $b \geq 0$ and $b \neq 0$. Therefore, the solution $x = \frac{y}{z}$ is well defined. By the Charnes-Cooper transformation (see, e.g., Chapter 4.3.2 of [8]), we have that $x = \frac{y}{z}$ is optimal for the relaxed WDP. The result follows because the solution is extremal for the relaxed WDP since the Charnes-Cooper transformation provides a one-to-one correspondence between extreme points. $\square$

B Missing proofs of Section 5.1, Section 5.2, and Section 5.3

PROOF OF LEMMA 5.2. We prove this lemma by contradiction. Suppose there exist j and j' such that $\sigma^*(j) > \sigma^*(j')$ and $\sum_{i \in [n]} v_i x_{ij} > \sum_{i \in [n]} v_i x_{ij'}$. It is easy to see that we can find j_1 and j_2 with $\sigma^*(j') \leq \sigma^*(j_1) = \sigma^*(j_2) - 1 < \sigma^*(j)$, and $\sum_{i \in [n]} v_i x_{ij_1} < \sum_{i \in [n]} v_i x_{ij_2}$.

Now consider a different permutation σ' which is the same as σ^* except $\sigma'(j_1) = \sigma^*(j_2)$ and $\sigma'(j_2) = \sigma^*(j_1)$. Note that $x \in X$ is an integral solution, and let i_1 be the advertiser matched with j_1 and i_2 be the advertiser matched with j_2 .

We have $v_{i_1} < v_{i_2}$, and we now compare the welfare of two permutations:

$$\begin{aligned} \text{Welfare}(x, \sigma^*) - \text{Welfare}(x, \sigma') &= v_{i_1} \cdot (\pi_{i_1}(x, \sigma^*) - \pi_{i_1}(x, \sigma')) + v_{i_2} \cdot (\pi_{i_2}(x, \sigma^*) - \pi_{i_2}(x, \sigma')) \\ &= \prod_{j': \sigma^*(j') < \sigma^*(j_1)} (1 - \sum_{i' \in [n]} x_{i'j'} p_{i'j'}) (v_{i_1} p_{i_1 j_1} (1 - (1 - p_{i_2 j_2})) - v_{i_2} p_{i_2 j_2} (1 - (1 - p_{i_1 j_1}))) \\ &= \prod_{j': \sigma^*(j') < \sigma^*(j_1)} (1 - \sum_{i' \in [n]} x_{i'j'} p_{i'j'}) p_{i_1 j_1} p_{i_2 j_2} (v_{i_1} - v_{i_2}) < 0. \end{aligned}$$

We get $\text{Welfare}(x, \sigma^*) < \text{Welfare}(x, \sigma')$ and this contradicts with σ^* maximizing Welfare. $\square$

PROOF OF LEMMA 5.3. We prove by induction on n' . The base case $n' = 0$ is trivial.

Assume the claim is true for $n' = k$. Now consider the case $n' = k + 1$. First of all, since $\sum_{i=1}^k \pi_i^{\text{restricted}}(x) \leq 1$, by the definition of $\pi_{k+1}^{\text{restricted}}(x)$, we know $\pi_{k+1}^{\text{restricted}}(x) \geq 0$.

We also have $\sum_{i=1}^{k+1} \pi_i^{\text{restricted}}(x) = \sum_{i=1}^k \pi_i^{\text{restricted}}(x) + \pi_{k+1}^{\text{restricted}}(x) \leq \sum_{i=1}^k \pi_i^{\text{restricted}}(x) + (1 - \sum_{i=1}^k \pi_i^{\text{restricted}}(x)) = 1$. For $\sum_{i=1}^{k+1} \pi_i^{\text{restricted}}(x)$ versus $\sum_{i=1}^{k+1} \pi_i(x)$, there are two cases:

- Case 1: $\sum_{i=1}^{k+1} \pi_i^{\text{restricted}}(x) = 1$. In this case, we just need to show $\sum_{i=1}^{k+1} \pi_i(x) \leq 1$. Since $1 - \sum_{i=1}^k \pi_i(x) \geq 0$, by definition, we have $\pi_{k+1}(x) \leq 1 \cdot (1 - \sum_{i=1}^k \pi_i(x))$. And this implies $\sum_{i=1}^{k+1} \pi_i(x) \leq 1$.
- Case 2: $\sum_{i=1}^{k+1} \pi_i^{\text{restricted}}(x) < 1$. In this case, we know $\pi_{k+1}^{\text{restricted}}(x) = \sum_{j \in [m]} x_{ij} \cdot p_{ij} \geq \pi_{k+1}(x)$. By induction, we also have $\sum_{i=1}^k \pi_i(x) \leq \sum_{i=1}^k \pi_i^{\text{restricted}}(x)$. Put them together, we get $\sum_{i=1}^{k+1} \pi_i(x) \leq \sum_{i=1}^{k+1} \pi_i^{\text{restricted}}(x)$. $\square$

PROOF OF LEMMA 5.4. Let s be the last item with cumulative non-cascade click-through rates at most 1, i.e. we have $\sum_{i=1}^s \pi_i^{\text{restricted}}(x) \leq 1$ and, either $s = n$ or $\sum_{i=1}^{s+1} \pi_i^{\text{restricted}}(x) > 1$. And it is easy to see that by definition for $i \in [s]$, $\pi_i^{\text{restricted}}(x) \geq \pi_i(x)$. For notation convenience, define $v_{n+1} = 0$ (it's for the case when we need to use v_{s+1} and $s = n$).

$$\begin{aligned} \text{Welfare}(x) &= \sum_{i \in [n]} v_i \pi_i(x) = \sum_{i=1}^s v_i \pi_i(x) + \sum_{i=s+1}^n v_i \pi_i(x) \leq \sum_{i=1}^s v_i \pi_i^{\text{restricted}}(x) - \sum_{i=1}^s v_{s+1} (\pi_i^{\text{restricted}}(x) - \pi_i(x)) + \sum_{i=s+1}^n v_{s+1} \pi_i(x) \\ &\leq \sum_{i=1}^s v_i \pi_i^{\text{restricted}}(x) + v_{s+1} \cdot \left(\sum_{i \in [n]} \pi_i(x) - \sum_{i=1}^s \pi_i^{\text{restricted}}(x) \right) \leq \sum_{i=1}^s v_i \pi_i^{\text{restricted}}(x) + v_{s+1} \cdot \left(\sum_{i \in [n]} \pi_i^{\text{restricted}}(x) - \sum_{i=1}^s \pi_i^{\text{restricted}}(x) \right) \\ &\leq \text{Welfare}^{\text{restricted}}(x). \end{aligned}$$

$\square$

PROOF OF LEMMA 5.5. Let s be the last item with cumulative non-cascade click-through rates at most 1/2, i.e. we have $\sum_{i=1}^s \pi_i^{\text{restricted}}(x) \leq 1/2$ and, either $s = n$ or $\sum_{i=1}^{s+1} \pi_i^{\text{restricted}}(x) > 1/2$.

And it is easy to see that by definition for $i \in [\min(n, s+1)]$, $\pi_i^{\text{restricted}}(x) \leq 2 \cdot \pi_i(x)$. For notation convenience, define $v_i = 0$ for $i > n$.

If $s = n$, we simply have $\sum_{i=1}^{s+1} v_i \pi_i^{\text{restricted}}(x) = \text{Welfare}^{\text{restricted}}(x) \geq \text{Welfare}^{\text{restricted}}(x)/2$. If $s < n$, we have $\sum_{i=1}^{s+1} v_i \pi_i^{\text{restricted}}(x) \geq v_{s+1} \cdot \frac{1}{2} \geq \sum_{i=s+2}^n v_i \pi_i^{\text{restricted}}(x) = \text{Welfare}^{\text{restricted}}(x) - \sum_{i=1}^{s+1} v_i \pi_i^{\text{restricted}}(x)$. And we get $\sum_{i=1}^{s+1} v_i \pi_i^{\text{restricted}}(x) \geq \text{Welfare}^{\text{restricted}}(x)/2$.

Then we have, $\text{Welfare}(x) = \sum_{i \in [n]} v_i \pi_i(x) \geq \sum_{i=1}^{s+1} v_i \pi_i(x) \geq \sum_{i=1}^{s+1} v_i \pi_i^{\text{restricted}}(x)/2 \geq \text{Welfare}^{\text{restricted}}(x)/4$. $\square$

PROOF OF LEMMA 5.8. We prove by contradiction. Suppose both k and k' satisfy the condition for being a discounted advertiser. Wlog let $k < k'$. Since $\pi_k^{\text{restricted}}(x) < \sum_{j \in [m]} x_{kj} p_{kj}$, by definition, we know $\pi_k^{\text{restricted}}(x) = 1 - \sum_{i=1}^{k-1} \pi_i^{\text{restricted}}(x)$. So $\sum_{i=1}^k \pi_i^{\text{restricted}}(x) = 1$. Now we have $\pi_{k'}^{\text{restricted}} \leq 1 - \sum_{i=1}^{k'-1} \pi_i^{\text{restricted}}(x) \leq 1 - \sum_{i=1}^k \pi_i^{\text{restricted}}(x) = 0$, and this gives a contradiction. $\square$

PROOF OF LEMMA 5.10. First of all, $\pi_i^{\text{restricted}}(x) = \pi_i^{\text{restricted}}(x')$ is straightforward to check by the definition using an induction on i . Then we have $\text{Welfare}^{\text{restricted}}(x) = \sum_{i=1}^n v_i \pi_i^{\text{restricted}}(x) = \sum_{i=1}^n v_i \pi_i^{\text{restricted}}(x') = \text{Welfare}^{\text{restricted}}(x')$.

For the second part of the lemma's claim, let us consider any advertiser i that is not a discounted advertiser in x . By definition, we know either $\pi_i^{\text{restricted}}(x) = 0$ or $0 < \pi_i^{\text{restricted}}(x) = \pi_i^{\text{budgeted}}(x)$.

In the first case where $\pi_i^{\text{restricted}}(x) = 0$, we know $x'_{ij} = 0$ for $j \in [m]$, and therefore $\pi_i^{\text{budgeted}}(x') = 0 = \pi_i^{\text{restricted}}(x) = \pi_i^{\text{restricted}}(x')$.

In the second case where $0 < \pi_i^{\text{restricted}}(x) = \pi_i^{\text{budgeted}}(x)$, we know $x'_{ij} = x_{ij}$ for $j \in [m]$, therefore $\pi_i^{\text{budgeted}}(x') = \pi_i^{\text{budgeted}}(x) = \pi_i^{\text{restricted}}(x) = \pi_i^{\text{restricted}}(x')$. $\square$

PROOF OF LEMMA 5.11. For any $\ell \in [n]$, by definition, we have that $\pi_\ell^{\text{restricted}}(x) = \sum_{j \in [m]} x_{\ell j} \cdot \min(p_{\ell j}, 1 - \sum_{i < \ell} \pi_i^{\text{restricted}}(x)) = \min(\sum_{j \in [m]} x_{\ell j} \cdot p_{\ell j}, 1 - \sum_{i < \ell} \pi_i^{\text{restricted}}(x)) = \min(\pi_\ell^{\text{budgeted}}(x), 1 - \sum_{i < \ell} \pi_i^{\text{restricted}}(x))$. Now we prove by induction that for any $\ell \in [n]$, $\sum_{i=1}^\ell \pi_i^{\text{restricted}}(x) \geq \sum_{i=1}^\ell \pi_i^{\text{budgeted}-k, \alpha}(x)$. The base case with $\ell = 0$ is trivial. Suppose the claim is true for $\ell - 1$, for the claim with ℓ , there are two cases:

- Case 1: $\sum_{i=1}^\ell \pi_i^{\text{restricted}}(x) = 1$. In this case, we simply have $\sum_{i=1}^\ell \pi_i^{\text{restricted}}(x) = 1 \geq \sum_{i=1}^\ell \pi_i^{\text{budgeted}-k, \alpha}(x) \geq \sum_{i=1}^\ell \pi_i^{\text{budgeted}-k, \alpha}(x)$.
- Case 2: $\sum_{i=1}^\ell \pi_i^{\text{restricted}}(x) \neq 1$. In this case, we know $\pi_\ell^{\text{restricted}}(x) \neq 1 - \sum_{i < \ell} \pi_i^{\text{restricted}}(x)$. Together with $\pi_\ell^{\text{restricted}}(x) = \min(\pi_\ell^{\text{budgeted}}(x), 1 - \sum_{i < \ell} \pi_i^{\text{restricted}}(x))$, we know $\pi_\ell^{\text{restricted}}(x) = \pi_\ell^{\text{budgeted}}(x)$. By induction hypothesis, we also have that $\sum_{i=1}^{\ell-1} \pi_i^{\text{restricted}}(x) \geq \sum_{i=1}^{\ell-1} \pi_i^{\text{budgeted}-k, \alpha}(x)$. Put them together, we get $\sum_{i=1}^\ell \pi_i^{\text{restricted}}(x) \geq \sum_{i=1}^\ell \pi_i^{\text{budgeted}-k, \alpha}(x)$.

Finally (here we set $v_{n+1} = 0$ for notation convenience), $\text{Welfare}^{\text{budgeted}-k, \alpha}(x) = \sum_{i=1}^n v_i \cdot \pi_i^{\text{budgeted}-k, \alpha}(x) = \sum_{\ell=1}^n (v_\ell - v_{\ell+1}) \cdot \sum_{i=1}^\ell \pi_i^{\text{budgeted}-k, \alpha}(x) \leq \sum_{\ell=1}^n (v_\ell - v_{\ell+1}) \cdot \sum_{i=1}^\ell \pi_i^{\text{restricted}}(x) = \sum_{i=1}^n v_i \cdot \pi_i^{\text{restricted}}(x) = \text{Welfare}^{\text{restricted}}(x)$. $\square$

PROOF OF THEOREM 5.12. The running time guarantee is straightforward to check. The algorithm has $O(n/\epsilon)$ iterations, and inside each iteration, both the budget matching algorithm and $\text{Welfare}^{\text{restricted}}$ computation can be done in time polynomial in n and m .

For the approximation guarantee, we denote the optimal allocation for $\text{Welfare}^{\text{restricted}}$ as x^* . By Lemma 5.8, x^* has at most one discounted advertiser. If x^* does have one discounted advertiser, let k to be that one, and set $\alpha = \lfloor \frac{2\pi_k^{\text{restricted}}(x^*)}{\pi_k^{\text{budgeted}}(x^*)\epsilon} \rfloor \cdot \epsilon/2$. If x^* does not have a discounted advertiser, let k be any arbitrary advertiser and set $\alpha = 1$.

Set $x' = \text{ZS}(x^*)$ using ZS defined in Definition 5.9. By Lemma 5.10, we know $\pi_i^{\text{restricted}}(x') = \pi_i^{\text{restricted}}(x^*)$ for $i \in [n]$, $\text{Welfare}^{\text{restricted}}(x') = \text{Welfare}^{\text{restricted}}(x^*)$, and $\pi_i^{\text{restricted}}(x') = \pi_i^{\text{budgeted}}(x')$ for $i \in [n] \setminus \{k\}$. For $\pi_k^{\text{budgeted}}(x')$, there are two cases depending on whether k is a discounted advertiser in x^* :

- If k is not a discounted advertiser in x^* , we know we set $\alpha = 1$. By Lemma 5.10, we have $\pi_k^{\text{restricted}}(x') = \pi_k^{\text{budgeted}}(x') = \alpha \cdot \pi_k^{\text{budgeted}}(x')$.
- If k is a discounted advertiser in x^* , we know $\pi_k^{\text{restricted}}(x^*) > 0$ and by the definition of ZS, $x'_{kj} = x^*_{kj}$ for $j \in [m]$. Therefore,

$$\pi_k^{\text{budgeted}}(x^*) = \pi_k^{\text{budgeted}}(x'). \text{ We get } \frac{2\pi_k^{\text{restricted}}(x^*)}{\pi_k^{\text{budgeted}}(x^*)\epsilon} \cdot \frac{\epsilon}{2} \cdot \pi_k^{\text{budgeted}}(x') = \frac{\pi_k^{\text{restricted}}(x')}{\pi_k^{\text{budgeted}}(x')} \cdot \pi_k^{\text{budgeted}}(x') = \pi_k^{\text{restricted}}(x'). \text{ Notice that } \alpha \leq \frac{2\pi_k^{\text{restricted}}(x^*)}{\pi_k^{\text{budgeted}}(x^*)\epsilon} \cdot \frac{\epsilon}{2} \leq \alpha + \epsilon/2. \text{ We get } \alpha \cdot \pi_k^{\text{budgeted}}(x') \leq \pi_k^{\text{restricted}}(x') \leq (\alpha + \epsilon/2) \cdot \pi_k^{\text{budgeted}}(x')$$

And we can conclude in both cases we have $\alpha \cdot \pi_k^{\text{budgeted}}(x') \leq \pi_k^{\text{restricted}}(x') \leq (\alpha + \epsilon/2) \cdot \pi_k^{\text{budgeted}}(x')$.

We then compare $\text{Welfare}^{\text{restricted}}(x')$ with $\text{Welfare}^{\text{budgeted}-k, \alpha}(x')$:

$$\begin{aligned} \text{Welfare}^{\text{budgeted}-k, \alpha}(x') &= \sum_{i \in [n] \setminus \{k\}} v_i \cdot \pi_i^{\text{budgeted}-k, \alpha}(x') + v_k \cdot \pi_k^{\text{budgeted}-k, \alpha}(x') \\ &= \sum_{i \in [n] \setminus \{k\}} v_i \cdot \pi_i^{\text{budgeted}}(x') + v_k \cdot \pi_k^{\text{budgeted}-k, \alpha}(x') = \sum_{i \in [n] \setminus \{k\}} v_i \cdot \pi_i^{\text{restricted}}(x') + v_k \cdot \pi_k^{\text{budgeted}-k, \alpha}(x') \\ &= \sum_{i \in [n] \setminus \{k\}} v_i \cdot \pi_i^{\text{restricted}}(x') + v_k \cdot \alpha \cdot \pi_k^{\text{budgeted}}(x') \geq \sum_{i \in [n] \setminus \{k\}} v_i \cdot \pi_i^{\text{restricted}}(x') + v_k \cdot \pi_k^{\text{restricted}}(x') - (\epsilon/2) \cdot v_k \cdot \pi_k^{\text{budgeted}}(x') \\ &\geq \sum_{i \in [n] \setminus \{k\}} v_i \cdot \pi_i^{\text{restricted}}(x') + v_k \cdot \pi_k^{\text{restricted}}(x') - (\epsilon/2) \cdot \text{Welfare}^{\text{restricted}}(x') = (1 - \epsilon/2) \cdot \text{Welfare}^{\text{restricted}}(x'). \end{aligned}$$

The inequality in the second last step comes from the fact that when we only allocate to advertiser k according to x'_k , the clickthrough rates don't need to be capped by 1 and the restricted welfare should be $v_k \sum_{j \in [m]} x'_{kj} p_{kj} = v_k \cdot \pi_k^{\text{budgeted}}(x')$ which is at most the optimal restricted welfare $\text{Welfare}^{\text{restricted}}(x^*) = \text{Welfare}^{\text{restricted}}(x')$.

Now we show $\mathbf{x}'$ is a valid solution for $\text{Welfare}^{\text{budgeted}-k,\alpha}$, i.e. $\mathbf{x}'$ satisfies constraint $\sum_{i=1}^n \pi_i^{\text{budgeted}-k,\alpha}(\mathbf{x}') \leq 1$. We have

$$\begin{aligned} \sum_{i=1}^n \pi_i^{\text{budgeted}-k,\alpha}(\mathbf{x}') &= \sum_{i \in [n] \setminus \{k\}} \pi_i^{\text{budgeted}-k,\alpha}(\mathbf{x}') + \pi_k^{\text{budgeted}-k,\alpha}(\mathbf{x}') = \sum_{i \in [n] \setminus \{k\}} \pi_i^{\text{budgeted}}(\mathbf{x}') + \pi_k^{\text{budgeted}-k,\alpha}(\mathbf{x}') \\ &= \sum_{i \in [n] \setminus \{k\}} \pi_i^{\text{restricted}}(\mathbf{x}') + \pi_k^{\text{budgeted}-k,\alpha}(\mathbf{x}') = \sum_{i \in [n] \setminus \{k\}} \pi_i^{\text{restricted}}(\mathbf{x}') + \alpha \cdot \pi_k^{\text{budgeted}}(\mathbf{x}') \\ &\leq \sum_{i \in [n] \setminus \{k\}} \pi_i^{\text{restricted}}(\mathbf{x}') + \pi_k^{\text{restricted}}(\mathbf{x}') \stackrel{\text{Lemma 5.3}}{\leq} 1 \end{aligned}$$

Since $\mathbf{x}'$ is a valid solution for $\text{Welfare}^{\text{budgeted}-k,\alpha}$, by the approximation guarantee of the PTAS for the budgeted matching problem, we know $\text{Welfare}^{\text{budgeted}-k,\alpha}(\mathbf{x}') \leq \text{Welfare}^{\text{budgeted}-k,\alpha}(\mathbf{x}^{k,\alpha})/(1 - \varepsilon/2)$.

Finally, by applying Lemma 5.11, we have $\text{Welfare}^{\text{budgeted}-k,\alpha}(\mathbf{x}^{k,\alpha}) \leq \text{Welfare}^{\text{restricted}}(\mathbf{x}^{k,\alpha})$.

To sum up, we have $\text{Welfare}^{\text{restricted}}(\mathbf{x}^*) = \text{Welfare}^{\text{restricted}}(\mathbf{x}') \leq \text{Welfare}^{\text{budgeted}-k,\alpha}(\mathbf{x}')/(1 - \varepsilon/2) \leq \text{Welfare}^{\text{budgeted}-k,\alpha}(\mathbf{x}^{k,\alpha})/(1 - \varepsilon/2)^2 \leq \text{Welfare}^{\text{restricted}}(\mathbf{x}^{k,\alpha})/(1 - \varepsilon/2)^2 \leq \text{Welfare}^{\text{restricted}}(\mathbf{x}^{k,\alpha})/(1 - \varepsilon)$.

Note that the output of Algorithm 1 has $\text{Welfare}^{\text{restricted}}$ at least $\text{Welfare}^{\text{restricted}}(\mathbf{x}^{k,\alpha})$. This completes the proof of the approximation guarantee of Algorithm 1. $\square$

C Missing proofs of Section 5.4

In this section we give the missing proofs in the analysis of the monotonic $O(\ln m)$ -approximation algorithm for welfare maximization.

Lemma 5.15. *If we can find a β -approximately optimal welfare allocation in each subgraph, then the algorithm of picking a bucket $\ell \in [1, \log_2(4m)]$ uniformly at random and using the β -approximately optimal allocation of G_ℓ would have an expected $\beta \cdot \log_2(4m)$ approximation ratio to the optimal welfare in the full graph. Furthermore, if the algorithm to find the allocation in each subgraph is monotonic, the overall algorithm (i.e., with the random picking) is also monotonic.*

Due to space limit, we will omit all proofs in this section. Readers are referred to the full version for the complete proofs.

As noted, we work with the base click-through rate when we find a matching in our allocation, and recall the corresponding definitions:

$$\text{Welfare}^{\text{base}}(\mathbf{x}) = \sum_{i \in [n]} v_i \pi_i^{\text{base}}(\mathbf{x}), \text{ where } \pi_i^{\text{base}}(\mathbf{x}) = \sum_{j \in [m]} x_{ij} \cdot p_{ij}.$$

As noted, (for the monotonicity of the allocation algorithm) we may use a permutation σ of the matched positions that can be different from the optimal permutation which ranks the positions in decreasing order of their matched advertisers' value. To keep things clear, in our analysis we explicitly write any allocation as a pair $(\mathbf{x}, \sigma)$ where $\mathbf{x}$ specifies the matching between advertisers and positions, and σ gives the ranking of the matched positions. We also use σ_0 to denote the (generic) optimal permutation which (when paired with any matching $\mathbf{x}$), ranks the matched positions by their matched advertiser's value. Using these notations, we can restate Lemma 5.4 as $\text{Welfare}^{\text{restricted}}(\mathbf{x}, \sigma_0) \geq \text{Welfare}(\mathbf{x}, \sigma_0)$ for any matching $\mathbf{x}$. Furthermore, since $\pi_i^{\text{base}}(\mathbf{x})$ is without any truncation or cascading effect, it's straightforward to see that for any matching and permutation $(\mathbf{x}, \sigma)$

$$\text{Welfare}^{\text{base}}(\mathbf{x}, \sigma) = \text{Welfare}^{\text{base}}(\mathbf{x}, \sigma_0) \geq \text{Welfare}^{\text{restricted}}(\mathbf{x}, \sigma_0) \geq \text{Welfare}(\mathbf{x}, \sigma_0). \quad (4)$$

Similar to the budgeted matching problem (as a proxy to optimize the restricted welfare), we consider a cardinality constraint where in subgraph G_ℓ we optimize over the space of matchings in G_ℓ with at most 2^ℓ edges (i.e. $\sum_{i,j} x_{ij} \leq 2^\ell$). Maximizing restricted welfare over matchings in G_ℓ remain (effectively) the same with or without the cardinality constraint, and together with Lemma 5.4 we have

Lemma C.1. *Let $\mathcal{X}_\ell$ be the space of feasible matchings in G_ℓ , and denote $|\mathbf{x}|$ of a matching $\mathbf{x}$ as the number of matched edges. For any matching $\mathbf{x}$ and permutation σ we have $\max_{\mathbf{x} \in \mathcal{X}_\ell, |\mathbf{x}| \leq 2^\ell} \text{Welfare}^{\text{base}}(\mathbf{x}, \sigma) \geq \max_{\mathbf{x} \in \mathcal{X}_\ell} \text{Welfare}(\mathbf{x}, \sigma_0)$. Note σ_0 is the (generic) permutation that orders the positions by their matched advertisers' values, and is the optimal permutation (w.r.t any matching) when maximizing Welfare.*

We use the (canonical) greedy strategy (Algorithm 2) to find a matching $\mathbf{x}_\ell$ (with $|\mathbf{x}_\ell| \leq 2^\ell$) and an associated permutation σ_ℓ in G_ℓ . We need to show that $\mathbf{x}_\ell$ gives a good approximation, and this largely follows the classic result that a greedy maximal matching is a 2-approximation of the maximum weight matching, with slight adaptation to accommodate the cardinality constraint.

Lemma C.2. *Let $(\mathbf{x}_\ell, \sigma_\ell)$ be the result of Algorithm 2 on G_ℓ , we have $\text{Welfare}^{\text{base}}(\mathbf{x}_\ell) \geq \frac{1}{2} \cdot \max_{\mathbf{x} \in \mathcal{X}_\ell, |\mathbf{x}| \leq 2^\ell} \text{Welfare}^{\text{base}}(\mathbf{x})$. Note here we omit the permutation since $\text{Welfare}^{\text{base}}$ is independent of it.*

We proceed with the following result to bound the gap between $\text{Welfare}(\mathbf{x}_\ell, \sigma_\ell)$ and $\text{Welfare}^{\text{base}}(\mathbf{x}_\ell, \sigma_\ell)$, which is analogous to Lemma 5.5.

Lemma C.3. *Let $(\mathbf{x}_\ell, \sigma_\ell)$ be the result of Algorithm 2 on G_ℓ , we have $\text{Welfare}(\mathbf{x}_\ell, \sigma_\ell) \geq \frac{1}{14} \cdot \text{Welfare}^{\text{base}}(\mathbf{x}_\ell)$.*

We are now ready to prove the main result of the greedy algorithm.

Lemma 5.16. *For any subgraph G_ℓ , Algorithm 2 (in polynomial time) finds an allocation of welfare at least $\frac{1}{28}$ of the optimal allocation in G_ℓ , and the allocation algorithm is monotonic.*